Topology

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1 Topology (拓樸學)

I Topological space (拓樸空間)

A topological space consists of a set X and a topology $\mathcal{T}$ on it, denoted as $(X, \mathcal{T})$. Where the set X is the set of points in the space, and the topology $\mathcal{T}$ is the set of subsets of X that satisfies:

1. $\varnothing, X \in \mathcal{T}$
2. Closed under arbitrary unions: $\forall \mathcal{S} \subseteq \mathcal{T} : \bigcup\limits_{A \in \mathcal{S}} A \in \mathcal{T}$
3. Closed under finite intersection: $\forall \mathcal{S} \subseteq \mathcal{T}$ s.t. $|\mathcal{S}| < \infty : \bigcap\limits_{A \in \mathcal{S}} A \in \mathcal{T}$

II Homeomorphism (同胚) or isomorphism (同構)

Topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ are called homeomorphic if there exists a mapping $f : X \to Y$ between them such that f is bijective and continuous and f^{-1} is continuous, written as $X \cong Y$, and f is called a homeomorphism between them.

III Open set (開集)

In a topological space $(X, \mathcal{T})$, an open subset $O \subseteq X$ is defined to be

$$O \in \mathcal{T}.$$

IV Open neighborhood (開鄰域)

An open neighborhood of a point P in a topological space X is any open subset $O \subseteq X$ such that $P \in O$.

An open neighborhood of a subset V in a topological space X is any open subset $O \subseteq X$ such that $V \subseteq O$.

V Interior (內部) of subset

The interior of a subset S of a topological space X, denoted by $\operatorname{int}_X S$, $\operatorname{int} S$, $\operatorname{int}_X(S)$, $\operatorname{int}(S)$, or S°, is defined as

$$\operatorname{int}(S) = \{x \in S \mid \exists \text{ open } U \subseteq X \text{ s.t. } x \in U \subseteq S\}.$$

VI Neighborhood (鄰域)

A subset U is a neighborhood of a point P if and only if there exists an open set O such that $P \in O \subseteq U$.

A subset U is a neighborhood of a subset V if and only if $V \subseteq \operatorname{int}(U)$.

VII Subspace topology

The subspace topology of a topological space $(X, \mathcal{T})$ in a subset I of X is defined as the set

$$\mathcal{I} = \{U \cap I | U \in \mathcal{T}\}.$$

VIII Topological embedding

A topological embedding is a homeomorphism onto its image with subspace topology.

IX Limit point (極限點), cluster point, or accumulation point (集積點)

Let S be a subset of a topological space X. A point x in X is a limit point, cluster point, or accumulation point of the set S if every neighborhood of x contains at least one point of S different from x itself.

X Closure (閉包)

The closure of a subset S of points in a topological space consists of all points in S together with all limit points of S.

XI Closed set (閉集)

A subset A of a topological space $(X, \mathcal{T})$ is called closed if its complement $X \setminus A \in \mathcal{T}$.

XII Borel set (博雷爾集)

A Borel set B is any set in a topological space $(X, \mathcal{T})$ that can be formed from open sets through the operations of countable union, countable intersection, and relative complement, that is,

1. Every element of $\mathcal{T}$ is a Borel set.
2. For any indexed family $(B_i)_{i=1}^{\infty}$ of Borel sets, $\bigcap_{i=1}^{\infty} B_i$ is a Borel set.
3. If B, C are Borel sets, then $B \setminus C$ is a Borel set.
4. Any other set is not a Borel set.

XIII Filter

A filter on a set X is a family $\mathcal{B}$ of subsets of X such that:

1. $X \in \mathcal{B}$.
2. $\varnothing \in \mathcal{B}$.
3. $A \in \mathcal{B} \land B \in \mathcal{B} \implies A \cap B \in \mathcal{B}$.
4. $A \subseteq B \subseteq X \land A \in \mathcal{A} \in \mathcal{B} \implies B \in \mathcal{B}$.

XIV Base or basis (基)

Given a topological space $(X, \mathcal{T})$, a base (or basis) for the topology $\mathcal{T}$ (also called a base for X if the topology is understood) is a family $\mathcal{B} \subseteq \mathcal{T}$ of open sets such that every open set of the topology can be represented as the union of some subfamily of $\mathcal{B}$.

The topology generated by a base $\mathcal{B}$, generally denoted by $\tau(\mathcal{B})$ can be defined to be follows: A subset $O \subseteq X$ is to be declared as open, if for all $x \in O$, there exists some $B \in \mathcal{B}$ such that $x \in B \subseteq O$.

XV Prefilter or filter base

$\mathcal{B}$ is called a prefilter if its upward closure $\uparrow \mathcal{B}$ is a filter.

XVI Quotient space (商空間)

Let X be a topological space and $\cong$ be an equivalence relation on it. The quotient space Y of X by $\cong$ is the quotient set of X by $\cong$ equipped with the quotient topology, defined as $U \subseteq Y$ is open in the quotient topology on Y if and only if $\{x \in X \mid [x] \in U\}$ is open in X and $V \subseteq Y$ is closed in the quotient topology on Y if and only if $\{x \in X \mid [x] \in V\}$ is closed in X.

XVII Type of topological spaces

i Distinct

Point x and point y in a topological space are distinct if and only if they are not the same point.

ii Kolmogorov space (柯爾莫果洛夫空間) or T$_0$ space

Two points in a topological space are topological distinguishable if and only if there exists an open set that contains one of them but not the other.

A topological space is called a Kolmogorov space or T$_0$ space if and only if any two distinct points in it are topological distinguishable. This condition is called the T$_0$ condition or the first separation theorem.

iii Kolmogorov quotient (space)

The Kolmogorov quotient (space) of a topological space is the quotient space of it under the equivalence relation topological indistinguishablity.

iv Fréchet space (弗蘭歇空間), accessible space, or T$_1$ space

Two points x and y in a topological space are separated if and only if there exists open sets U, V such that $x \in U \wedge y \notin U \wedge y \in V \wedge x \notin V$.

A topological space is called a Fréchet space, accessible space, or T$_1$ space if and only if any two distinct points in it are separated. This condition is called the T$_1$ condition or the second separation theorem. A topological space is a T_1 space implies it is a T_0 space.

v Symmetric space or R_0 space

A topological space is called a symmetric space or R_0 space if and only if any two topological distinguishable points in it are separated.

vi Hausdorff space (郝斯多夫空間), separated space (分離空間), or T_2 space

Points x and y in a topological space are separated by neighborhoods if and only if there exists a neighborhood U of x and a neighborhood V of y such that U and V are disjointed.

A topological space is called a Hausdorff space, separated space, or T_2 space if and only if any two distinct points in it are separated by neighborhoods. This condition is called the T_2 condition or the third separation theorem. A topological space is a T_2 space implies it is a T_1 space.

vii Urysohn space or $\mathrm{T}_{2½}$ space

Points x and y in a topological space are separated by closed neighborhoods if and only if there exists a closed neighborhood U of x and a closed neighborhood V of y such that U and V are disjointed.

A topological space is called a Urysohn space or $\mathrm{T}_{2½}$ space if and only if any two distinct points in it are separated by closed neighborhoods. A topological space is a $\mathrm{T}_{2½}$ space implies it is a T_2 space.

viii Completely Hausdorff space, functionally Hausdorff space, or completely T_2 space

Points x and y in a topological space X are separated by a (continuous) function if and only if there exists a continuous function $f: X \to [0,1]$ such that $f(x) = 0$ and $f(y) = 1$.

A topological space is called a completely Hausdorff space, functionally Hausdorff space, or completely T_2 space if and only if any two distinct points in it are separated by a continuous function. A topological space is a completely T_2 space implies it is a $\mathrm{T}_{2½}$ space.

ix Regular space (正則空間)

A topological space X is called a regular space if and only if for any closed set F and any point $x \notin F$, there exists a neighbourhood U of x and a neighbourhood V of F such that U and V are disjointed.

x Regular Hausdorff space or T_3 space

A topological space is called a regular Hausdorff space or T_3 space if and only if it is a regular space and a Hausdorff space.

xi Completely regular space

A closed set F and a point $x \notin F$ in a topological space X are separated by a (continuous) function if and only if there exists a continuous function $f: X \to [0,1]$ such that $f(x) = 0$ and

$F \subseteq f^{-1}(\{1\})$.

A topological space X is called a completely regular space if and only if any closed set F and any point $x \notin F$ in it are separated by a (continuous) function. A topological space is a completely regular space implies it is a regular space.

xii Completely regular Hausdorff space, Tychonoff space, completely T_3 space, or $T_{3½}$ space

A topological space X is called a completely regular Hausdorff space, Tychonoff space, completely T_3 space, or $T_{3½}$ space if and only if it is a completely regular space and a Hausdorff space. A topological space is a completely T_3 space implies it is a completely T_2 space.

xiii Normal space (正規空間)

Disjoint closed sets E and F in a topological space are separated by neighborhoods if and only if there exists a neighborhood U of E and a neighborhood V of F such that U and V are disjointed.

A topological space is called a normal space if and only if any two distinct closed sets in it are separated by neighborhoods. A topological space is a normal space implies it is a regular space.

xiv Urysohn's lemma

Two closed sets E and F in a topological space X are separated by a (continuous) function if and only if there exists a continuous function $f: X \to [0,1]$ such that $\subseteq f^{-1}(\{0\})$ and $F \subseteq f^{-1}(\{1\})$.

The Urysohn's lemma states that, a topological space is a normal space if and only if any two disjoint closed sets in it are separated by a (continuous) function.

xv Normal Hausdorff space or T_4 space

A topological space is called a normal Hausdorff space or T_4 space if and only if it is a normal space and a T_1 space. A topological space is a normal Hausdorff space implies it is a regular Hausdorff space.

xvi Completely normal space or hereditarily normal space

A topological space is called a completely normal space or hereditarily normal space if and only if any subspace of it is a normal space.

xvii Completely normal Hausdorff space, T_5 space, or completely T_4 space

A topological space is called a completely normal Hausdorff space, T_5 space, or completely T_4 space if and only if it is a completely normal space and a T_1 space. A topological space is a completely normal Hausdorff space implies it is a normal Hausdorff space and a completely regular Hausdorff space.

xviii Perfectly normal space

Two closed sets E and F in a topological space X are precisely separated by a (continuous) function if and only if there exists a continuous function $f: X \to [0,1]$ such that $E = f^{-1}(\{0\})$ and $F = f^{-1}(\{1\})$.

A topological space is called a perfectly normal space if and only if any two disjoint closed sets in it are precisely separated by a (continuous) function. A topological space is a perfect normal space implies it is a completely normal space.

xix Perfectly normal Hausdorff space, T_6 space, or perfectly T_4 space

A topological space is called a perfectly normal Hausdorff space, T_6 space, or perfectly T_4 space if and only if it is a perfectly normal space and a T_1 space. A topological space is a perfectly normal Hausdorff space implies it is a completely normal Hausdorff space.

xx Connected space (連通空間)

A topological space is called disconnected if and only if it can be expressed as the union of two disjoint non-empty open sets; otherwise, it is called connected.

xxi Path-connected space (路徑連通空間)

A topological space X is called path-connected if and only if for any two points $x, y \in X$, there existes a continuous function $f: [0,1] \to X$ such that $f(0) = x$ and $f(1) = y$. A topological space is path-connected implies it is connected.

xxii Simply connected space or 1-connected space (單連通空間)

A topological space is called simply connected if and only if it is path-connected, and that for any two points $x, y \in X$ and any two continuous functions $f: [0,1] \to X$ and $g: [0,1] \to X$ such that $f(0) = g(0) = x$ and $f(1) = g(1) = y$, there exists a continuous function $F: [0,1] \times [0,1] \to X$ such that $F(x,0) = f(x)$ and $F(x,1) = g(x)$.

xxiii Compact (緊緻) (subset)

A subset K of a topological space X is said to be compact if for every collection C of open subsets of X such that

$$K \subseteq \bigcup_{S \in C} S,$$

called a open cover, there is a finite subcollection $F \subseteq C$ such that

$$K \subseteq \bigcup_{S \in F} S,$$

called a finite subcover.

xxiv Compact space

A topological space X is called compact iff for every collection C of open subsets of X such that

$$X = \bigcup_{S \in C} S,$$

called a open cover, there is a finite subcollection $F \subseteq C$ such that

$$X = \bigcup_{S \in F} S,$$

called a finite subcover.

xxv (Weakly) locally compact space

A topological space X is called (weakly) locally compact iff every point in it has a compact neighbourhood.

xxvi Strongly locally compact or locally relatively compact space

A topological space X is called strongly locally compact or locally relatively compact iff every point in it has a closed compact neighbourhood.

A Hausdorff space is weakly locally compact iff it is strongly locally compact.

xxvii σ-compact space

A topological space X is called σ-compact iff its topology is the union of countably many compact spaces.

xxviii σ-locally compact space

A topological space X is called σ-locally compact iff it is σ-compact and locally compact.

xxix Second-countable space or completely separable space

A topological space is second-countable iff it has a countable basis.

XVIII Locally finite cover

Let X be a topological space, and $\{U_i \mid i \in I\}$ be a cover of X. We say that U is locally finite if for all $x \in X$, there exists $V \ni x$ that is open in X such that the set $\{i \in I \mid V \cap U_i \neq \varnothing\}$ is finite.

XIX Dense (稠密) subset

A subset A of a topological space X is said to be a dense subset of X if for every $x \in X$ and every neighborhood U of x, $U \cap A \neq \varnothing$.

XX Comparison of topologies

Let τ_1 and τ_2 be two topologies on a set X.

τ_1 is said to be a coarser (weaker or smaller) topology than τ_2, and τ_2 is said to be a finer (stronger or larger) topology than τ_1, iff τ_1 is a subset of τ_2.

τ_1 is said to be a strictly coarser topology than τ_2, and τ_2 is said to be a strictly finer topology than τ_1, iff τ_1 is a proper subset of τ_2.

The binary relation $\subseteq$ defines a partial ordering relation on the set of all possible topologies on X.

XXI Product topology or Tychonoff topology

The product topology or Tychonoff topology on a set $X = \prod_{i=1}^{n} X_i$ where X_i are topological spaces and $n > 1$ is defined as the coarsest topology on X such that all ith projection map are continuous. X with the product topology on X is called a product space. Unless otherwise specified, product topology is assumed on a Cartesian product of topological spaces.

XXII Metric space (度量空間 or 賦距空間)

i Metric space

Metric space is an ordered pair (M, d) where M is a set and d is a metric on M, i.e., a function $d : M \times M \to \mathbb{R}$ satisfying the following axioms for all points $x, y, z \in M$:

1. The distance from a point to itself is zero: $d(x, x) = 0$.
2. (Positivity) The distance between two distinct points is always positive:$x \neq y \implies d(x, y) > 0$.
3. (Symmetry) The distance from x to y is always the same as the distance from y to x: $d(x, y) = d(y, x)$.
4. The triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

For a normed space $(X, \|\cdot\|)$, unless otherwise specified, it is equipped with a metric d defined as

$$d(x, y) = \|x - y\|, \quad \forall x, y \in X.$$

ii Cauchy sequence (柯西序列)

Given a metric space (x, d), a sequence of elements of X:

$$x_1, x_2, x_3, \ldots$$

is Cauchy if for every positive real number ε there is a positive integer N such that for all positive integers $m, n > N$:

$$d\left(x_m, x_n\right) < \varepsilon.$$

iii Complete metric space (完備度量空間) or Cauchy space (柯西空間)

A metric space (X, d) is complete if every Cauchy sequence of points in X has a limit that is also in X.

iv Complete subset or Cauchy subset

A subset S of a metric space (X, d) is complete if every Cauchy sequence of points in S has a limit that is also in S.

v (Open) ball (開球)

In a metric space (X, d), the open ball or ball of radius $r > 0$ and centered at the point $a \in X$, denoted as $B_{<r}(a)$, $B_r(a)$, or $B(a, r)$, is defined as:

$$B_r(a) \coloneqq \{p \in X \,:\, d(a, p) < r\}.$$

vi Closed ball (閉球)

In a metric space (X, d), the closed ball of radius $r \geq 0$ and centered at the point $a \in X$, denoted as $B_{\leq r}(a)$, is defined as:

$$B_{\leq r}(a) \coloneqq \{p \in X \,:\, d(a, p) \leq r\}.$$

vii Bounded (有界) subset

A subset S of a metric space (M, d) is called bounded if and only if there exists $r > 0$ such that for all $s, t \in S$, we have $d(s, t) < r$.

viii Totally bounded or precompact subset

A subset S of a metric space (M, d) is called totally bounded or precompact if and only if there eixsts $r > 0$ such that there exists a finite collection of open balls of radius r that is a cover of S.

ix ε-net

A subset E of a metric space X is an ε-net for another subset A of X if

$$A \subseteq \bigcup_{x \in E} B(x, \varepsilon).$$

If a finite ε-net exists for every $\varepsilon > 0$, then A is totally bounded.

x A compact subset of a Hausdorff space is closed

Statement: If a subset K of a Hausdorff space X is compact, then it must be closed.

Proof. We want to show that K is closed, that is, $X \setminus K$ is open.

For any $a \in X \setminus K$, for each $x \in K$, by the definition of Hausdorff space, there exist a disjoint open set U_x such that $x \in U_x$, $a \in V_x$, and $U_x \cap V_x = \varnothing$.

The collection of $C = \{U_x \mid x \in K\}$ is an open cover of K. Since K is compact, a finite subset D of C is a cover of K.

Corresponding to each $U_i \in D$, we have an open neighborhood V_i of a such that $U_i \cap V_i = \varnothing$. Let $V = \bigcap_{U_i \in D} V_i$. Then V is an open neighborhood of a and is disjoint from $\bigcap_{U_i \in D} U_i \supseteq K$, so

$$V \cap K = \varnothing.$$

For every $a \in X \setminus K$, we found an open neighborhood V of a such that

$$V \subseteq X \setminus K \wedge V \cap K = \varnothing,$$

so K is closed. □

xi A closed subset of a compact set is compact

Statement: If K is a compact subset of a topological space X and $F \subseteq K$ is closed, then F is compact.

Proof. Let C be an open cover of F. Then

$$C \cup \{X \setminus F\}$$

is an open cover of K.

Since K is compact, there is a finite subcollection $D = C' \cup \{X \setminus F\}$ of $C \cup \{X \setminus F\}$ where $C' \subseteq C$ that is a cover of K. Then C' is a finite subcollection of C that is a cover of F. Thus F is compact. □

xii A compact subset of a metric space is bounded

Statement: If K is a compact subset of a metric space (X, d), then K is bounded.

Proof. Since K is compact, for any point $p \in X$, there exist a finite collection of open balls

$$\{B(p, n) \mid n \in \mathbb{N} \wedge n \leq m \in \mathbb{N}\}$$

that is a cover of K, then $K \subseteq B(p, m)$, so K is bounded. □

xiii A complete subset of a complete metric space is closed

Statement: If a subset A of a complete metric space is compact, them A is closed.

Proof. Since a metric space is necessarily a Hausdorff space, any convergent sequence must have a unique limit. Since A is complete, every Cauchy sequence in A converges to a point in A. Thus, A is closed. □

xiv A closed subset of a complete metric space is complete

Statement: If a subset A of a complete metric space X is closed, them A is compact.

Proof. Let (x_n) be a Cauchy sequence in A. Since the space is complete, it converges to some $x \in X$. Since A is closed in X, any limit of a convergent sequence in A must be in A. Thus, A is complete. □

xv A closed and bounded subset of an Euclidean space is compact

Statement: If a subset A of an Euclidean space $\mathbb{R}^n$ is closed and bounded, then A is compact.

Proof. Let (x_k) be any sequence in A. We want to show that it has a convergent subsequence with its limit in A.

Since A is bounded, there exists $M > 0$ such that

$$\|x_k\| \leq M, \quad \forall k.$$

So

$$x_k \in B(0, M), \quad \forall k.$$

By the Bolzano–Weierstrass theorem, there exists a subsequence (x_{k_j}) that coverges to some point $x \in \mathbb{R}^n$.

Since A is closed, it is complete. Thus $x \in A$. □

xvi Heine–Borel Theorem (海涅-博雷爾定理)

For a subset S of an Euclidean space $\mathbb{R}^n$, the following two statements are equivalent:

- S is compact;
- S is closed and bounded.

xvii A complete and totally bounded subset of a complete metric space is compact

Statement: If a subset A of a complete metric space (x, d) is complete and totally bounded, then A it is compact.

Proof. Take any sequence (x_n) in A. We want to find a convergent subsequence whose limit lies in A.

For $\varepsilon = 1$, since A is totally bounded, there exists a finite cover of A by closed balls of radius 1:

$$A \subseteq \bigcup_{i=1}^{N_1} B_1(y_i).$$

Since (x_n) is infinite but the cover is finite, by the pigeonhole principle, one of these balls must contain infinitely many terms of (x_n). Denote this ball by $B_1(y_{i_1})$ and pick the subsequence $(x_n^{(1)})$ of (x_n) that is entirely in $B_1(y_{i_1})$.

Repeat it for $\varepsilon = \frac{1}{2}, \frac{1}{3}, \ldots$, we can construct a nested sequence of subsequences

$$(x_n^{(1)}) \supseteq (x_n^{(2)}) \supseteq \ldots$$

where $(x_n^{(k)})$ lies in a closed ball of radius $\frac{1}{k}$, $B_{\frac{1}{k}}(y_{i_k})$.

Pick the diagonal sequence $y_k = x_k^{(k)}$. Then

$$y_k \in (x_n^{(k)}) \subseteq B_{\frac{1}{k}}(y_{i_k}),$$

and the sequence (y_k) is a Cauchy sequence in A because

$$\forall N \in \mathbb{N}: \ m, n \geq N \implies d(y_m, y_n) \leq \frac{2}{N}.$$

Since A is complete, (y_k) must converges to some point $y \in A$. □

xviii Generalized/general Heine–Borel Theorem

For a subset S of a complete metric space (X, d), the following two statements are equivalent:

- S is compact;
- S is complete and totally bounded.

xix A totally bounded subset is bounded

Statement: If a subset A of a metric space (X, d) is totally bounded, them A is bounded.

Proof. By total boundedness, there exist finitely many points $x_1, x_2, \dots x_n \in X$ such that:

$$A \subseteq \bigcup_{i=1}^{n} B(x_i, 1).$$

Pick one of these centers, say x_1.

For any $a \in A$, there exists some $i \in \{1, 2, \dots n\}$ such that

$$d(a, x_i) < 1.$$

By triangle inequality:

$$d(a, x_1) \leq d(a, x_i) + d(x_i, x_1) \leq 1 + d(x_i, x_1).$$

Let:

$$M = 1 + \max_{i \in \{1,2,\dots n\}} d(x_i, x_1).$$

So, for all $a \in A$:

$$d(a, x_1) < M.$$

Thus:

$$A \subseteq B(x_1, M).$$

Therefore, A is bounded. □

xx All norms on a finite-dimensional vector space are equivalent

Statement: Let V be a finite-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ and $\dim V = n$. If $\|\cdot\|_a$ and $\|\cdot\|_b$ are two norm on V, then there exist constants $c, C > 0$ such that for all $v \in V$,

$$c\|v\|_a \leq \|v\|_b \leq C\|v\|_a,$$

called that the two norms are equivalent, and c, C are called the equivalence constants.

Proof. First consider $V = \mathbb{R}^n$. The set

$$S_a = \{x \in \mathbb{R}^n \mid \|x\|_a = 1\}$$

is closed and bounded in $\mathbb{R}^n$, hence compact.

Define

$$f(x) = \|x\|_b, \quad x \in S_a.$$

Since $\|\cdot\|_b$ is continuous, f is continuous on the compact set S_a. Therefore f attains a minimum and maximum on S_a. So there exist $m, M \geq 0$ such that

$$m \leq f(x) \leq M.$$

For any $v \neq 0$, write

$$u = \frac{v}{\|v\|_a} \in S_a.$$

Then

$$\|v\|_b = \|v\|_a \cdot \|u\|_b.$$

So

$$m\|v\|_a \leq \|v\|_b \leq M\|v\|_a.$$

Thus $c = m, C = M$ are the equivalence constants. □

xxi A bounded subset of a finite-dimensional Banach space is totally bounded

Statement: If a subset A of a n-dimensional Banach space X with $n \in \mathbb{N}$ is bounded, then A is totally bounded.

Proof. Pick a linear homeomorphism

$$T : \mathbb{R}^n \to X.$$

Define a norm on $\mathbb{R}^n$ by

$$\|v\|_* := \|Tv\|_X, \quad \forall v \in \mathbb{R}^n.$$

By that all norms on a finite-dimensional vector space are equivalent, the Euclidean norm $\|\cdot\|_2$ and $\|\cdot\|_*$ are equivalent, that is, there exist constants $c, C > 0$ such that for every $v \in \mathbb{R}^n$,

$$c\|v\|_2 \leq \|v\|_* \leq C\|v\|_2. \qquad (1)$$

Let $A \subseteq X$ be bounded. Then $B := T^{-1}(A)$ is bounded with respect to the metric induced by $\|\cdot\|_*$ and, by (1), is bounded with respect to the metric induced by $\|\cdot\|_2$. So there is $R > 0$ with $B_{\|\cdot\|_2}(0, R) \supseteq B$. For any $\varepsilon > 0$ and $\delta := \frac{\varepsilon}{C}$, there is a finite collection of $B_{\|\cdot\|_2}(v_i, R)$ where $v_i \in \mathbb{R}^n$ that is a cover of $B_{\|\cdot\|_2}(0, R)$.

Take any $b \in B$. Then $b \in B_{\|\cdot\|_2}(v_i, R)$ for some v_i, so $\|b - v_i\|_2 < \delta$.

By (1), we get

$$\|T(b) - T(v_i)\|_X = \|b - v_i\|_* \leq C\|b - v_i\|_2 < C\delta = \varepsilon.$$

Therefore the finite set of all $T(v_i)$ is an ε-net for A. Since $\varepsilon > 0$ was arbitrary, A is totally bounded. □

XXIII Net (網)

i Net (網)

A net in X, denoted as $x_\bullet = (x_a)_{a\in A}$ or $\langle x_a\rangle_{a\in A}$, is a function of the form $x_\bullet : A \to X$ whose domain A is some directed set, and whose values are $x_\bullet(a) = x_a$ in a topological space X. Elements of a net's domain are called its indices.

ii Eventually or residually

A net $x_\bullet = (x_a)_{a\in A}$ in X is said to be eventually or residually in a set S if there exists some $a \in A$ such that for every $b \in A$ with $b \geq a$, the point $x_b \in S$.

iii Limit of nets

A point x in a topological space X is called a limit point or limit of a net $x_\bullet$ in X if, for every open neighborhood U of x, the net $x_\bullet$ is eventually in U, also called that the net converges to/towards x, and denoted as $x_\bullet \to x$, $x_a \to x$, $\lim x_\bullet \to x$, $\lim_{a\in A} x_a \to x$, or $\lim_a x_a \to x$.

If $\lim x_\bullet \to x$ and this limit is unique (i.e. $\lim x_\bullet \to y$ only if $x = y$), then one writes $\lim x_\bullet = x$, $\lim_{a\in A} x_a = x$, or $\lim_a x_a = x$.

iv Cofinal (共尾) or frequent subset

A subset B of a directed set A is a cofinal or frequent subset if for all $a \in A$, there exists $b \in B$ such that $b \geq a$.

v Cofinally (共尾地) or frequently

A net $x_\bullet = (x_a)_{a\in A}$ in X is said to be cofinally or frequently in a set S if for every $a \in A$ there exists $b \in A$ such that $b \geq a \wedge x_b \in S$.

vi Cluster point, or accumulation point (集積點)

A point x in a topological space X is called a cluster point or accumulation point of a net $x_\bullet$ in X if, for every open neighborhood U of x, the net $x_\bullet$ is cofinally in U, also called that the net clusters to/towards x.

A limit point of a net is necessary a cluster point of it.

vii Subnet or Willard-subnet

A net $s_\bullet = (s_i)_{i\in I}$ is called a subnet or Willard-subnet of another net $x_\bullet = (x_a)_{a\in A}$ if there exists an order-preserving map $h : I \to A$ such that $h(I)$ is a cofinal subset of A and

$$s_i = x_{h(i)} \quad \forall i \in I.$$

viii Ultranet or universal net

A net $x_\bullet$ in X is called an ultranet or a universal net if for every subset $S \subseteq X$, $x_\bullet$ is eventually in S or $X \setminus S$.

A point is a limit point of a ultranet if and only if it a cluster point of it.

ix In Hausdorff spaces

A net in a Hausdorff space has at most one limit; if it has one, the limit point of it is the only cluster point of it.

Proof. TODO □

x Willard-subnet limit and cluster point theorems

For any nonempty net $\langle a_n \rangle_{n \in A}$ and its any Willard-subnet $\langle b_n \rangle_{n \in B}$, if x is a limit point of a_n, then x is a limit point of b_n.

Proof. Let $h : B \to A$ be order-preserving and $h(B)$ be a cofinal subset of A. Let U be an open neighborhood of x. There exists $a \in A$ such that $n \geq a \implies a_n \in U$. Then, for any $b \in B$ such that $h(b) \geq a$, $m \geq b \implies b_m = a_{h(m)} \in U$. □

For any nonempty net $\langle a_n \rangle_{n \in A}$ and its any Willard-subnet $\langle b_n \rangle_{n \in B}$, if x is a cluster point of b_n, then x is a cluster point of a_n.

Proof. Let $h : B \to A$ be order-preserving and $h(B)$ be a cofinal subset of A. Let U be an open neighborhood of x. For any $b \in B$, there exists $m \geq b$ such that $a_{h(m)} = b_m \in U$. Also, for any $c \in A$, there exists $d \in B$ such that $h(d) \geq c$. Thus, for any $a \in A$, there exists $n \geq a$ such that $a_n \in U$. □

Its special case when $A = B = \mathbb{N}_0$ or $\mathbb{N}_1$, and $a_n \in \mathbb{R}$ is called the subsequence convergence theorem.

XXIV Manifold

i Euclidean neighborhood

An Euclidean neighborhood of a point in a topological space is a neighborhood of it that is homeomorphic to an open subset of $\mathbb{R}^n$ for some $n \in \mathbb{N}_0$. Such Euclidean neighborhood is called n-dimensional.

A Euclidean neighborhood homeomorphic to an open ball in $\mathbb{R}^n$ is called a Euclidean ball.

ii Locally Euclidean space

A topological space X is called locally Euclidean iff it can be covered by n-dimensional Euclidean neighborhoods for some $n \in \mathbb{N}_0$. Such locally Euclidean space is called n-dimensional or an n-locally Euclidean space.

Euclidean balls form a basis for the topology of a locally Euclidean space.

iii (Topological) manifold (流形)

A (topological) manifold is a locally Euclidean space that is a Hausdorff space. An n-dimensional (topological) manifold or n-manifold is an n-dimensional locally Euclidean space that is a Hausdorff space.

An 1D manifold is called curve. A 2D manifold is a called a surface.

iv (Coordinate) chart

For any Euclidean neighborhood U, a homeomorphism $\phi\colon U \to \phi(U) \subseteq \mathbb{R}^n$ is called a (coordinate) chart on U.

v Atlas

For a locally Euclidean space M, an atlas $\{(U_i, \phi_i) \mid i \in I\}$ on M is a family of ordered pairs of Euclidean neighborhood U_i and coordinate chart ϕ_i on U_i such that $\{U_i \mid i \in I\}$ covers M.

vi Transition function or transition map

For an atlas $\{(U_i, \phi_i) \mid i \in I\}$ on an n-dimensional locally Euclidean space M, the transition function $\phi_i{\phi_j}^{-1}$ for $i, j \in I$ is defined as the function

$$\phi_i{\phi_j}^{-1}\colon \phi_i(U_i \cap U_j) \to \phi_j(U_i \cap U_j);\ x \mapsto \phi_i({\phi_j}^{-1}(x)).$$

Such a map is a homeomorphism between open subsets of $\mathbb{R}^n$.

vii Differentiable and smooth

An atlas $\{(U_i, \phi_i) \mid i \in I\}$ is called a differentiable atlas iff for all $i, j \in I$, the transition map $\phi_i{\phi_j}^{-1}$ is differentiable.

An atlas $\{(U_i, \phi_i) \mid i \in I\}$ is called a (n-times) continuously differentiable atlas iff for all $i, j \in I$, the transition map $\phi_i{\phi_j}^{-1}$ is (n-times) continuously differentiable.

An atlas $\{(U_i, \phi_i) \mid i \in I\}$ is called a C^k atlas iff for all $i, j \in I$, the transition map $\phi_i{\phi_j}^{-1}$ is a C^k function.

An atlas $\{(U_i, \phi_i) \mid i \in I\}$ is called a smooth (C^∞) atlas iff for all $i, j \in I$, the transition map $\phi_i{\phi_j}^{-1}$ is a smooth (C^∞) function.

A differentiable manifold is a manifold such that there exists a differentiable atlas on it.

A (n-times) continuously differentiable manifold is a manifold such that there exists a (n-times) continuously differentiable atlas on it.

A C^k manifold is a manifold such that there exists a C^k atlas on it.

A smooth (C^∞) manifold is a manifold such that there exists a smooth (C^∞) atlas on it.

Let M be an n-dimensional differentiable manifold with differentiable atlas $\{(U_i, \phi_i) \mid i \in I\}$. A map $f : M \to M$ is a differentiable map iff for any charts (U_i, ϕ_i) such that $p \in U_i \wedge i \in I$ and (U_j, ϕ_j) such that $f(p) \in U_j \wedge j \in I$,

$$\phi_j \circ f \circ \phi_i^{-1} : \phi_i(U_i \cap f^{-1}(U_j)) \subseteq \mathbb{R}^n \to \mathbb{R}^n$$

is differentiable.

Let M be an n-dimensional (n-times) continuously differentiable manifold with (n-times) continuously differentiable atlas $\{(U_i, \phi_i) \mid i \in I\}$. A map $f : M \to M$ is an (n-times) continuously differentiable map iff for any charts (U_i, ϕ_i) such that $p \in U_i \wedge i \in I$ and (U_j, ϕ_j) such that $f(p) \in U_j \wedge j \in I$,

$$\phi_j \circ f \circ \phi_i^{-1} : \phi_i(U_i \cap f^{-1}(U_j)) \subseteq \mathbb{R}^n \to \mathbb{R}^n$$

is (n-times) continuously differentiable.

Let M be an n-dimensional C^k manifold with C^k atlas $\{(U_i, \phi_i) \mid i \in I\}$. A map $f : M \to M$ is a C^k map iff for any charts (U_i, ϕ_i) such that $p \in U_i \wedge i \in I$ and (U_j, ϕ_j) such that $f(p) \in U_j \wedge j \in I$,

$$\phi_j \circ f \circ \phi_i^{-1} : \phi_i(U_i \cap f^{-1}(U_j)) \subseteq \mathbb{R}^n \to \mathbb{R}^n$$

is a C^k function.

Let M be an n-dimensional smooth (C^∞) manifold with smooth (C^∞) atlas $\{(U_i, \phi_i) \mid i \in I\}$. A map $f : M \to M$ is a smooth (C^∞) map iff for any charts (U_i, ϕ_i) such that $p \in U_i \wedge i \in I$ and (U_j, ϕ_j) such that $f(p) \in U_j \wedge j \in I$,

$$\phi_j \circ f \circ \phi_i^{-1} : \phi_i(U_i \cap f^{-1}(U_j)) \subseteq \mathbb{R}^n \to \mathbb{R}^n$$

is a smooth (C^∞) function.

viii Diffeomorphism

Given two differentiable manifolds X, Y, a continuously differentiable map $f : X \to Y$ is a diffeomorphism iff it is a bijection and its inverse is differentiable. If there exists a diffeomorphism between two differentiable manifolds, they are called diffeomorphic.

Given two C^k manifolds X, Y, a C^k map $f : X \to Y$ is a C^k-diffeomorphism iff it is a bijection and its inverse is C^k. If there exists a C^k-diffeomorphism between two C^k manifolds, they are called C^k-diffeomorphic.

ix Local diffeomorphism

Given two differentiable manifolds X, Y, a map $f : X \to Y$ is a local diffeomorphism iff for any point $x \in X$, there exists an open neighborhood U of X such that $f_U : U \to f(U)$ is a diffeomorphism. If there exists a local diffeomorphism between two differentiable manifolds, they are called locally diffeomorphic.

Given two C^k manifolds X, Y, a map $f : X \to Y$ is a local C^k-diffeomorphism iff for any point $x \in X$, there exists an open neighborhood U of X such that $f_U : U \to f(U)$ is a C^k-diffeomorphism. If there exists a local C^k-diffeomorphism between two differentiable manifolds, they are called locally C^k-diffeomorphic.

XXV Combination with abstract algebra

i Topological group (拓樸群)

A topological group is a triple $(X, \tau, *)$ such that (X, τ) is a topological space, $(X, *)$ is a group, and the maps of $(x, y) \in X^2 \mapsto x * y$ and $x \mapsto x^{-1}$ are continuous mapa. Such topology τ is called a group topology.

ii Topological field (拓樸域)

A topological field is defined with a topological space (X, τ) and a field $(X, +, *)$ such that the maps of $a \mapsto -a$ and $a \mapsto a^{-1}$ are continuous maps.

XXVI Discrete space

i Discrete topology

A discrete topology on X is a topology such that every subset of X is open.

ii Discrete (topological) space

A space equipped with discrete topology.

iii Discrete subspace

A topological subspace whose subspace topology is a discrete topology.